



wearable takes more time to charge. Despite improvement in batteries' capacity in the last decade, the stored charge drains rapidly after a limited period of time. This decreases a wearable's lifecycle and, as a result, the battery needs to be regularly replaced or changed to a rechargeable battery, which increases maintenance and cost.

Also, a BMS that lasts for only four to six years is not helpful. Battery lifetime is a critical issue in wearable devices.

Small size factor is another critical issue for wearable devices. A battery's complex internal structure can significantly increase the size and cost of the devices, making it inconvenient for people to wear them.

Some of the new ways to achieve efficient power management are:

1. Bluetooth LE 5, Wi-Fi, or RFID. They consume low power (in micro-watts) and offer effective data transfer facilities on an SoC.

2. Advanced CMOS technology. It has enabled the evolution of low-power circuits and systems. It allows the device to operate at a sub-microwatt power level.

3. Energy harvesting sources. With the shrinking transistor size, there is a need for efficient management of power sources. This means we should

consider generating energy from thermal, vibration, or solar sources. However, energy harvesting brings some design challenges with itself, since harvested energy has voltage swings and depends on the availability and size of the harvester.

Block diagram changes

A typical block diagram of wearable device has sensors for heart rate, ECG, pulse-rate, pedometer, and others. These invariably require good interfacing with the controller through some ADCs or some other type of analogue front-end converters. Right selection of the controller is important to save power.

If we consider some blocks such as blocks of the sensor, energy source, and power management (for analogue front-end processing, wireless transceivers, and processor), the energy source powers up the entire device. So, the power management block should efficiently distribute this power to the different blocks in the system.

After that, the sensor block will collect data and send it to the digital processor where the digitised data would be processed and saved in memory. The data can also be stored to calculate the average value. Finally, the wireless transceiver block sends

the processed data to another user/receiver.

These blocks generally require low-power electric current in micro-amperes (mA). So, most of the time, the low-power transmitter outputs something like 160mW at a maximum data rate of 200kbps, whereas it practically consumes around 190nW. When the device is duty-cycled by 0.013%, usually the wireless transmitter consumes a much larger amount of power and therefore operates in a duty-cycle fashion in order to minimise the overall power consumption of the device.

Patients admitted to an ICU are those who are critically ill and are generally unable to breathe on their own. So, to ventilate their lungs, the clinician creates an airway inside a patient's windpipe to provide artificial respiration. This is known as Intubation, wherein a plastic tube called an Endotracheal Tube is inserted into the mouth so that one end reaches the trachea (windpipe) while the other end is connected to a mechanical ventilator.

However, due to lung secretion, mucous, or blood clot, these tubes get obstructed, which results in compromised respiratory efficiency and can result in brain damage or even death. Since there are only two to three minutes available for taking proper action, the blockages need to be removed immediately for the endotracheal tube to pass air, else the patient will die within no time.

To perform intubation, expert hands are required, generally of an anesthesiologist. But in most cases, these specialists are not available in time. So, in such a scenario, a smart device can be worn by the patients in ICU or other critical hospital environments. The device's sensor can act as a connection between the mechanical ventilator and endotracheal tube to monitor and detect obstructions in real time.

After designing the smart devices with the help of traditional operational amplifiers, ultrasonic sensors, or acoustic sensors, the main challenge is the battery since endotracheal intubation can last four to eight days. To overcome this, you need to develop a circuit that enables the device to